

Unit 6 : Word Vector Representation

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Introduction to Word Embedding:

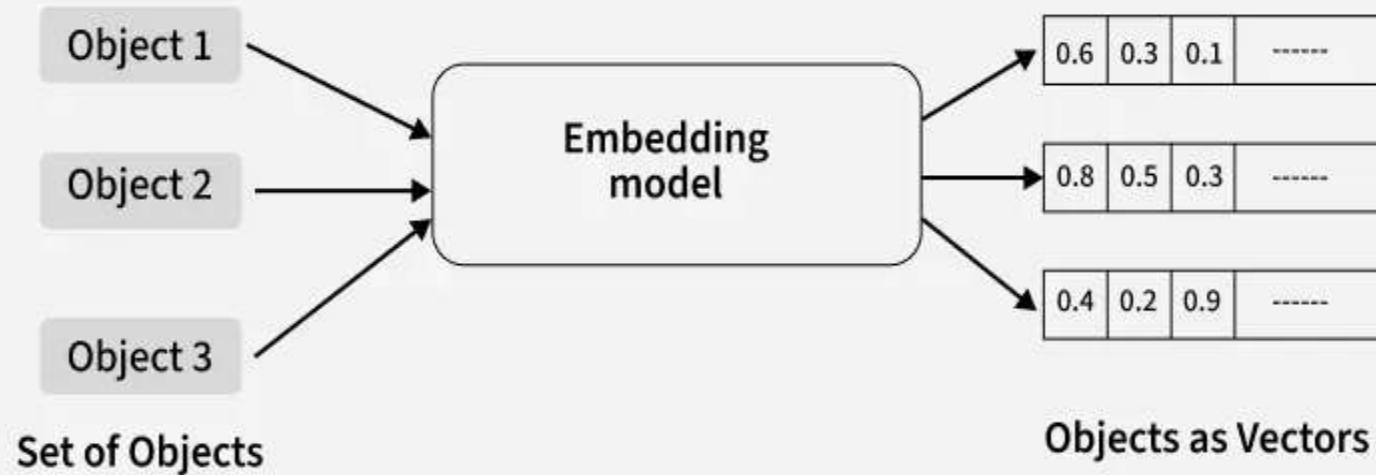
Word embedding is a machine learning technique in Natural Language Processing (NLP) that translates words or phrases into dense, multi-dimensional numerical vectors.

Over the years, several core architectures have been developed to generate word embeddings:

Word Embeddings are numeric representations of words in a lower-dimensional space, that capture semantic and syntactic information. They play a important role in Natural Language Processing (NLP) tasks. Here, we'll discuss some traditional and neural approaches used to implement Word Embeddings, such as TF-IDF, Word2Vec, and GloVe.

Introduction to Word Embedding:

Embeddings in Natural Language Processing



Neural Language Model

Neural language models (NLMs) in Natural Language Processing (NLP) are deep learning-based systems that learn to predict or generate text by analyzing language patterns, replacing traditional n-gram models with higher accuracy and context retention.

There are several components of the neural language models, they include:

Word embeddings

RNNS

Attention mechanism

Natural Language Processing (NLP) helps machines to understand and process human languages either in text or audio form. It is used across a variety of applications from speech recognition to language translation and text summarization.

Neural Language Model

Components of a Neural Language Model

1. Input Layer

Words are converted into vectors using embeddings.

Example:

Word	Vector Representation
------	-----------------------

cat	[0.2, 0.7, -0.1]
-----	------------------

dog	[0.1, 0.8, -0.2]
-----	------------------

Similar words get similar vectors

2. Hidden Layers

Neural networks learn relationships between words.

Common architectures:



Neural Language Model

Feedforward Neural Networks (FFNN)

Recurrent Neural Networks (RNN)

LSTM

GRU

Transformers

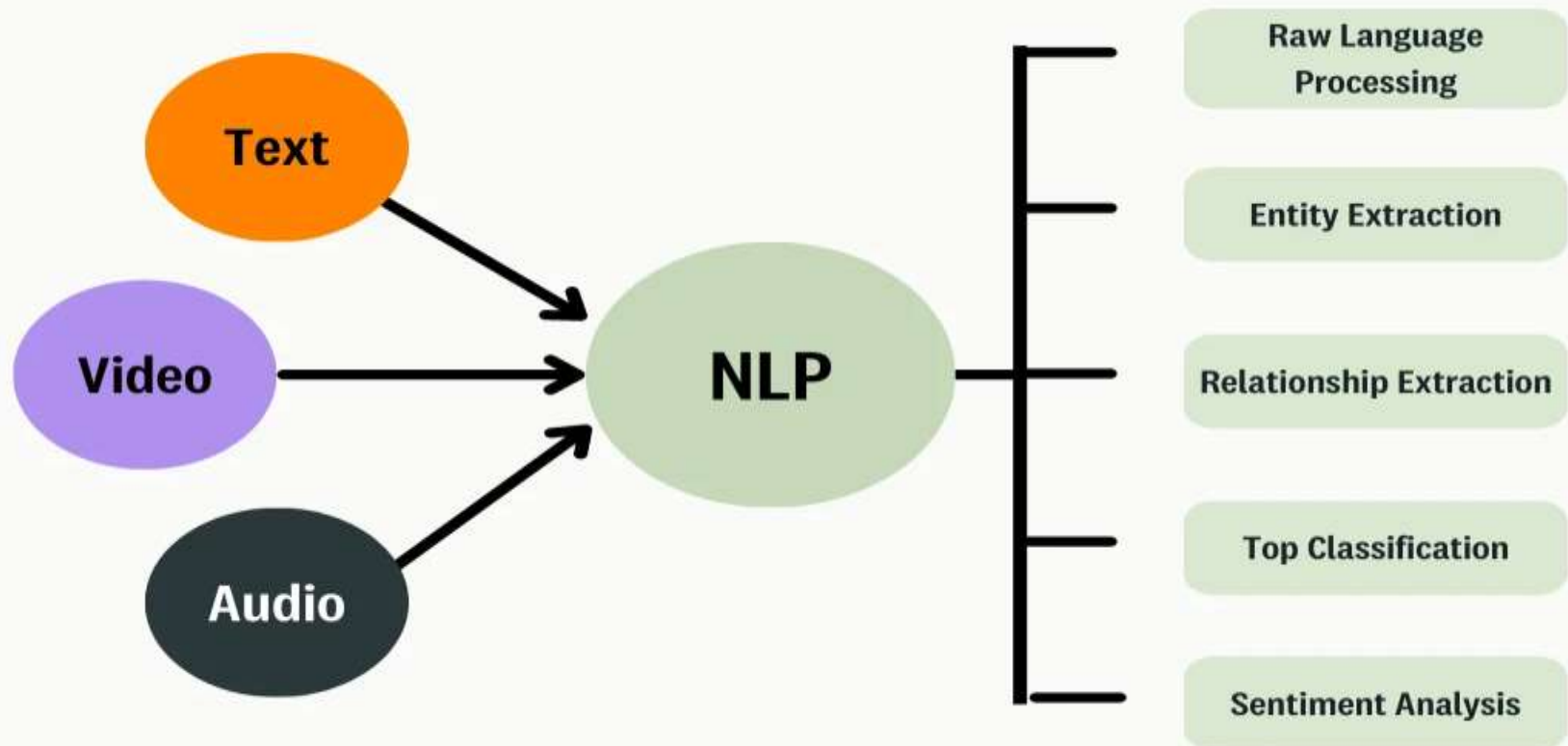
3. Output Layer

Predicts probability distribution for the next word.

Example:

Word	Probability
apple	0.60
mango	0.25
car	0.01

Neural Language Model

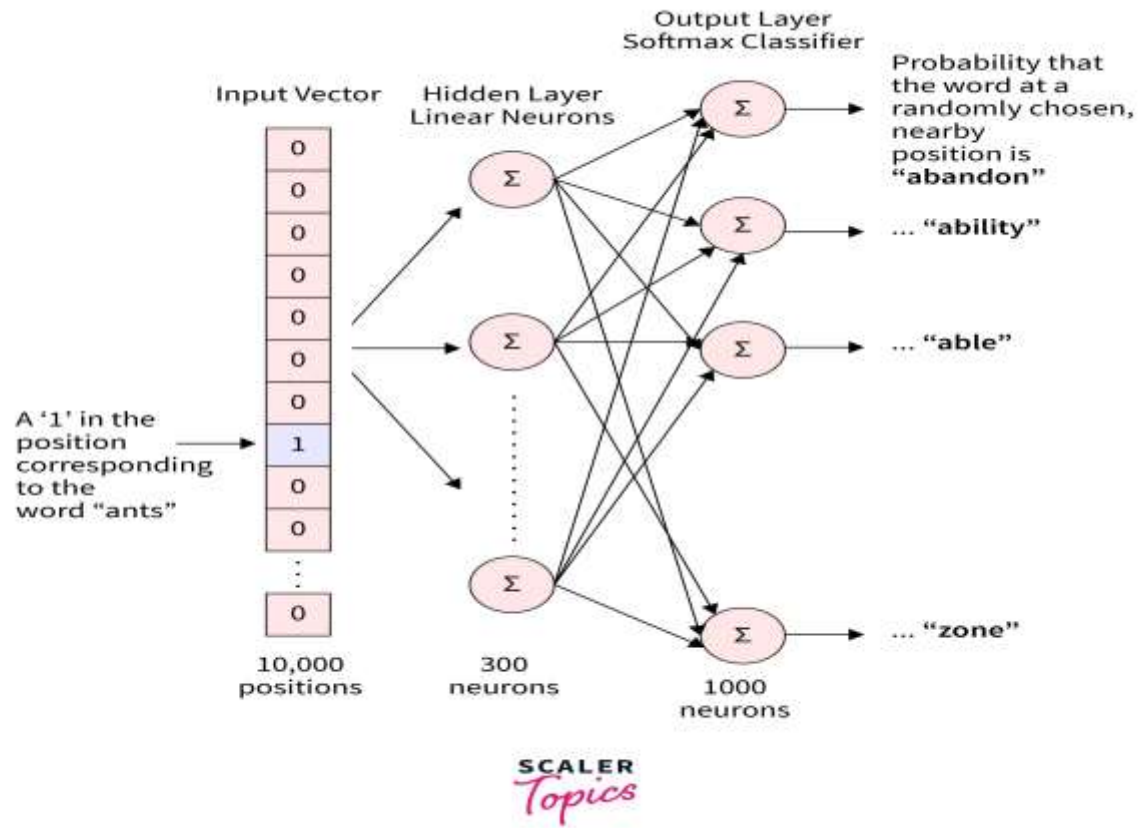


Word2Vec

Word2Vec is a foundational Natural Language Processing (NLP) technique developed by Google that converts words into numerical vectors. It maps words with similar meanings to similar coordinates in a multidimensional space, enabling computers to understand context, identify synonyms, and perform mathematical operations on words.

Word2Vec is a word embedding technique in NLP that represents words as vectors in a continuous space. Developed by Google, it captures semantic relationships by assigning similar vectors to words with similar meanings.

Word2Vec

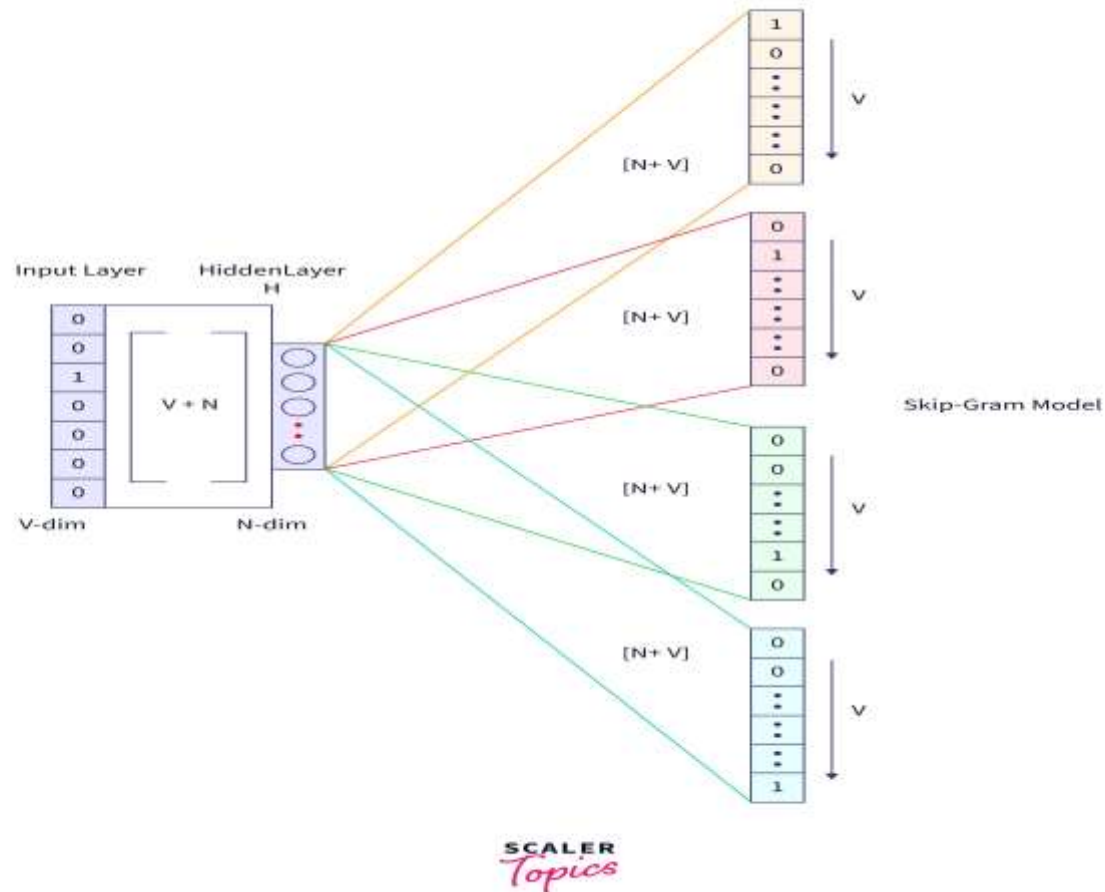


Skip-Gram Model

The skip-gram model was introduced by Mikolov et al. in their paper "Efficient Estimation of Word Representations in Vector Space" (2013). The skip-gram model is a way of teaching a computer to understand the meaning of words based on the context they are used in. An example would be training a computer to understand the word "dog" by looking at sentences where "dog" appears and seeing the words that come before and after it. By doing this, the computer will be able to understand how the word "dog" is commonly used and will be able to use that understanding in other ways.

In conclusion, the skip-gram model is a popular method for generating word embeddings, dense vector representations of words in a vocabulary.

Skip-Gram Model



Model Components

Natural Language Processing (NLP) models are AI systems designed to comprehend, interpret, and generate human language. They are built upon two primary processing branches—Natural Language Understanding (NLU) and Natural Language Generation (NLG)—and rely on core linguistic and architectural components.

The components of an NLP model work together to convert human language into numerical form, learn language patterns, and generate intelligent predictions or responses.

Architecture

The architecture of an NLP model describes how different components are organized and connected to process language data and generate predictions.

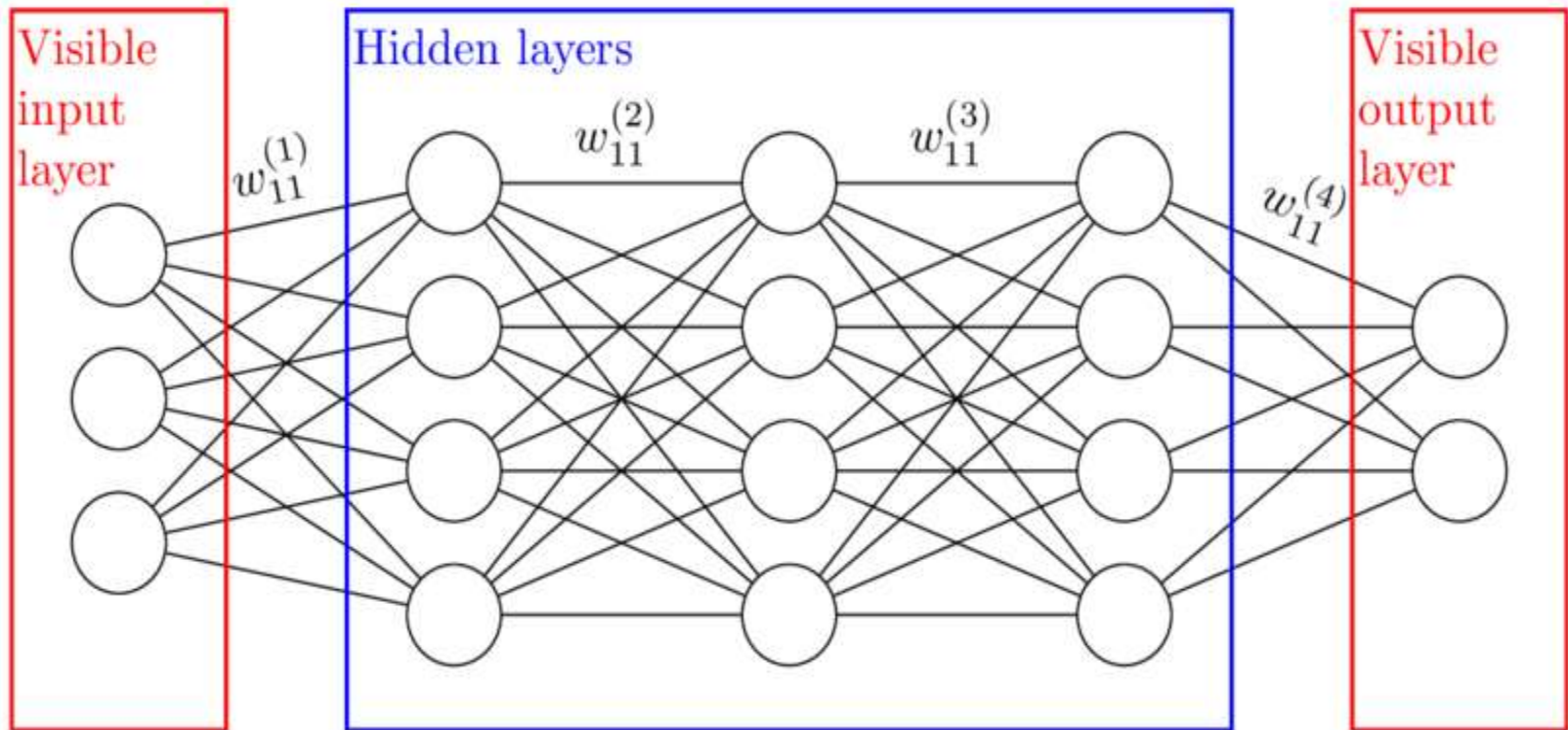
The architecture of a modern Natural Language Processing (NLP) model typically follows a structured pipeline that converts raw text into numerical data, processes it through complex neural layers, and generates a meaningful output.

Hidden Layer

Hidden layers in Natural Language Processing (NLP) are the intermediate computational layers between the input and output in a neural network. They process language by transforming raw text representations into complex, high-level features, allowing models to "understand" context, grammar, and semantics.

Hidden layers are the intermediate computational tiers in an artificial neural network situated between the input layer and the output layer. They act as the "brain" of deep learning, performing complex mathematical transformations that allow the AI to recognize intricate patterns and abstract features in data.

Hidden Layer



Output Layer

In Natural Language Processing (NLP), the output layer is the final layer of a neural network that converts computed features into actionable, human-interpretable linguistic results. Its structure and activation function entirely depend on the specific NLP task being performed.

Main Function of Output Layer

The output layer:

Receives processed features from hidden layers

Computes probabilities or scores

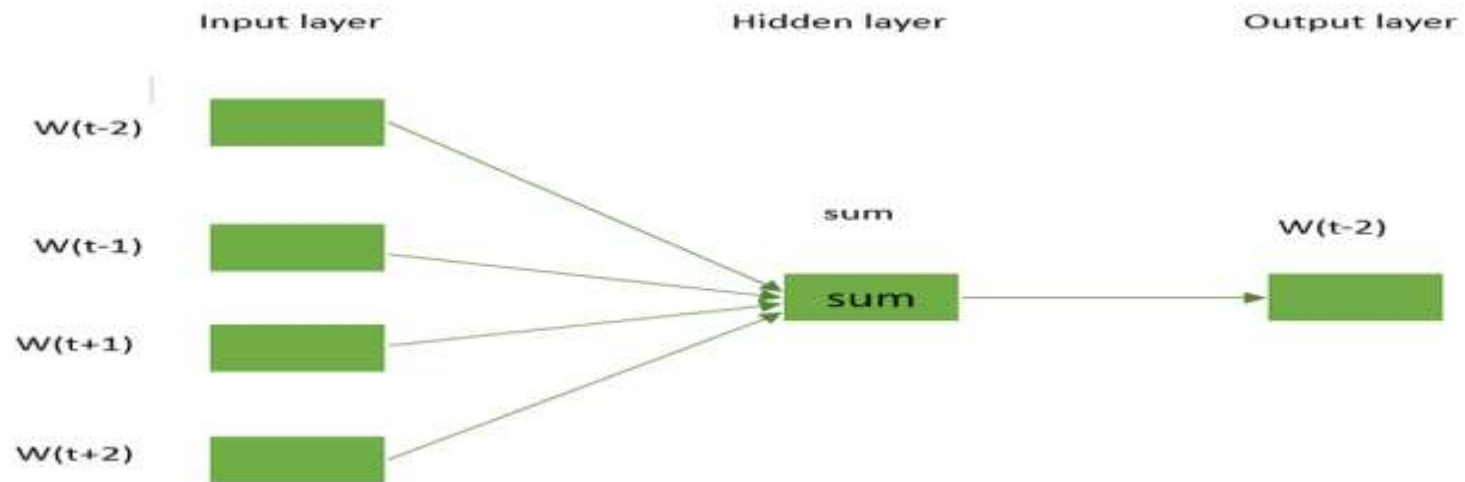
Produces final prediction

CBOW Model

Continuous Bag of Words (CBOW) is a popular natural language processing technique used to generate word embeddings. Word embeddings are important for many NLP tasks because they capture semantic and syntactic relationships between words in a language. CBOW is a neural network-based algorithm that predicts a target word given its surrounding context words. It is a type of "unsupervised" learning, meaning that it can learn from unlabeled data, and it is often used to pre-train word embeddings that can be used for various NLP tasks such as sentiment analysis, text classification, and machine translation.

Architecture of CBOW Model

The CBOW model uses the context words around the target word in order to predict it. Consider the above example "She is a great dancer." The CBOW model converts this phrase into pairs of context words and target words. The word pairings would appear like this ([she, a], is), ([is, great], a) ([a, dancer], great) having window size=2.



Subsampling Frequent Word

Subsampling in NLP is a technique used during the training of word embeddings (like Word2Vec) to randomly discard highly frequent words (e.g., "the", "in", "a"). This reduces dataset noise, speeds up training, and significantly improves the model's ability to learn meaningful representations for rarer, more informative words.

Mikolov et al. (the creators of Word2Vec) established a specific probability for discarding a word w_i

$$P(w_i) = 1 - t$$

Subsampling Frequent Word

$P(w_i)$ is the probability that the word w_i is discarded (not included in the training sample).

$f(w_i)$ is the frequency of the word w_i (number of occurrences of w_i divided by the total number of words in the corpus).

t is a chosen threshold parameter, typically around 10^{-5} .

Negative Sampling

Negative sampling is a computationally efficient training technique used in Natural Language Processing (NLP). Instead of calculating probabilities for every word in the vocabulary, it simplifies training by treating complex multiclass problems as a series of binary classification tasks, vastly reducing training time and memory usage.

When working with natural language processing (NLP) tasks, especially word embeddings, efficiently optimizing models like Word2Vec becomes critical due to the vast vocabulary sizes involved. Negative Sampling is a powerful technique that simplifies and accelerates the training of such models by approximating the computationally expensive softmax function.

Efficient Softmax approximations

Efficient approximations to the softmax function are widely used in modern NLP because the standard softmax becomes expensive when the vocabulary size is very large (e.g., 50k–1M tokens).

The bottleneck comes from computing:

$$P(w_i) = \frac{e^{z_i}}{\sum_{j=1}^V e^{z_j}}$$

Word embeddings → Negative Sampling

Huge vocabulary LM → Adaptive or Sampled Softmax

Interpretable sparse attention → Entmax/Sparsemax

Classic scalable LM → Hierarchical Softmax

Production transformers → optimized dense softmax + kernel tricks

Sampling-based approximations for Softmax

Sampling-based softmax approximations are techniques that avoid computing the full normalization over a huge vocabulary by using only a small subset of sampled words during training.

Instead of computing:

$$P(w_i) = \frac{e^{w_i \cdot v}}{\sum_{j=1}^V e^{w_j \cdot v}}$$

Main Sampling-Based Softmax Approximations

1. Negative Sampling

Introduced in:

Efficient Estimation of Word Representations in Vector Space

Used heavily in:

word2vec

Glove, Cross-lingual word embedding models

GloVe in NLP

GloVe: Global Vectors for Word Representation

GloVe (Global Vectors) is a word embedding model that combines:
global matrix factorization methods
local context window methods
to learn dense vector representations of words.

Properties of GloVe Embeddings

Semantic Regularities

Famous analogy behavior:

king-man+woman≈queen

king-man+woman≈queen

because linear relationships emerge naturally.

Glove, Cross-lingual word embedding models

Cross-Lingual Word Embeddings

Cross-lingual embeddings map words from multiple languages into a shared vector space.

Goal:

dogEnglish $\approx$ chienFrench dog English $\approx$ chien French

in embedding space.

An embedding model in NLP is a model that converts discrete items like words, sentences, or documents into dense numerical vectors in a continuous space.

The main goal is:

Similar meanings $\rightarrow$ nearby vectors.

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